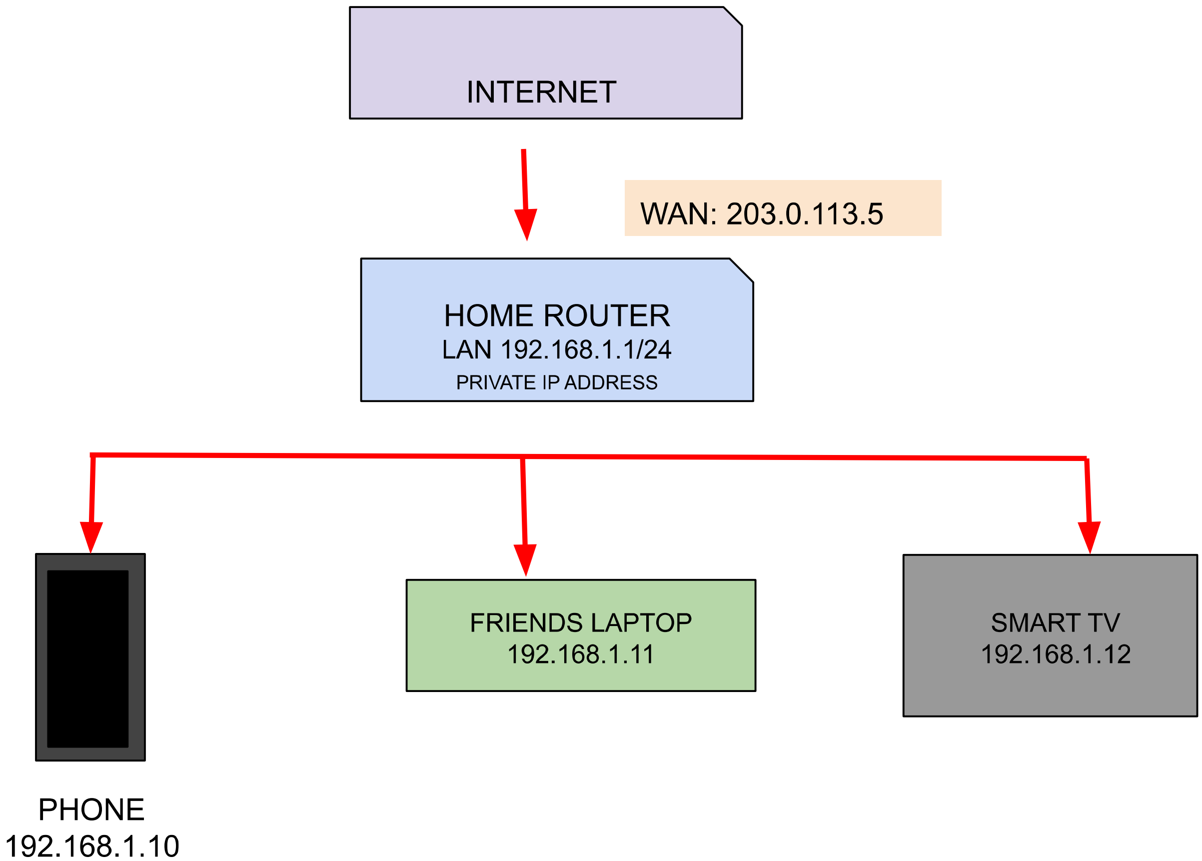
Task 1: Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.



Setup: All three devices sit on the router's LAN side, subnet `192.168.1.0/24`, mask `255.255.255.0`. The router's LAN interface (`192.168.1.1`) is the default gateway for all three.

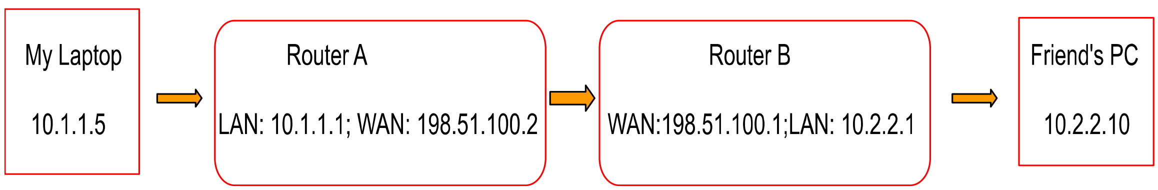
On the WAN side, the router gets a public IP (`203.0.113.5`, made up) from the ISP — via DHCP from their network.

Task 2: Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made. Hint: Use made-up IP addresses and show the routing table entries that would be checked at each hop.

Scenario: My laptop is on one subnet, my friend's computer is on a different subnet, and they're connected through two routers.

Addressing:

- Router A: LAN 10.1.1.1/24, WAN 198.51.100.1/30
    - My laptop: 10.1.1.5/24, gateway 10.1.1.1 (Router A, LAN interface)
- Router B: WAN 198.51.100.2/30, LAN 10.2.2.1/24
    - Friend's computer: 10.2.2.10/24, gateway 10.2.2.1 (Router B, LAN interface)



Source IP: 10.1.1.5 | Destination IP: 10.2.2.10

Step-by-step packet journey:

1. **My Laptop checks if the destination is local.** It compares 10.2.2.10 against its own subnet mask /24. Since 10.2.2.0 ≠ 10.1.1.0, the destination isn't on the local network.
2. **Laptop sends the packet to its default gateway** (Router A, 10.1.1.1), because it has no direct route to 10.2.2.0/24.
3. **Router A checks its routing table:**It matches 10.2.2.0/24 and forwards the packet out its WAN interface to 198.51.100.2 (Router B).

Destination	Mask	Next hop	Interface
10.1.1.0	/24	Directly connected	LAN
198.51.100.2	/30	Directly Connected	WAN
10.2.2.0	/24	198.51.100.2	WAN

4. **Router B receives the packet** on its WAN interface and checks its own table:

Destination	Mask	Next hop	Interface
10.2.2.0	/24	Directly connected	LAN
10.1.1.0	/24	198.51.100.1	WAN

10.2.2.0/24 is directly connected, so Router B forwards the packet straight out its LAN interface.

5. **Friend's computer receives the packet.** The reply follows the exact reverse path back through Router B → Router A → your laptop.

At every hop, the routing decision is the same logical step: compare the destination IP against each table entry's network/mask combination, pick the most specific (longest prefix) match, and forward accordingly. If nothing matches, the default route (0.0.0.0/0) is used as a last resort.

Task 3: Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

The corporate network utilizes a **Hub-and-Spoke topology**. The Central Hub (HQ/Cloud Core) acts as the Hub, while the regional city branches (Mumbai, Delhi, Bangalore) act as the Spokes.

Physical Connectivity: Each regional branch is connected to the Core Server infrastructure via dedicated, point-to-point **Leased Lines (WAN circuits).** These circuits terminate directly onto the Core Router's physical or modular WAN interfaces, isolating internal branch-to-core traffic entirely from the public Internet.

We will set up a **Central Operations Hub** that routes traffic to the three regional branches via

Here is how the network subnets are allocated:

- **Central Hub (HQ/Cloud Core):** 10.0.0.0/24
- **Mumbai Branch (Kitchens & Riders):** 10.1.0.0/24
- **Delhi Branch (Kitchens & Riders):** 10.2.0.0/24
- **Bangalore Branch (Kitchens & Riders):** 10.3.0.0/24

Destination Network	Purpose / Traffic Type	Subnet Mask	Next Hop IP	Out Interface
10.0.0.0	Central Order Databases & API Gateways	255.255.255.0 (/24)	Directly Connected	GigabitEthernet (LAN)
10.1.0.0 (LAN)	Mumbai Branch	255.255.255.0 (/24)	192.168.10.2 (WAN)	Virtual Serial0/0/0 (WAN - Mumbai Link)
10.2.0.0 (LAN)	Delhi Branch	255.255.255.0 (/24)	192.168.20.2 (WAN)	Virtual Serial0/0/1 (WAN - Delhi Link)
10.3.0.0 (LAN)	BangaloreBranch	255.255.255.0 (/24)	192.168.30.2 (WAN)	Virtual Serial0/1/0 (WAN - Bangalore Link)
0.0.0.0 ("Anywhere else / Everywhere else in the world.")	Public Cloud, Customer Apps & Payment Gateways	0.0.0.0 (/0)	203.0.113.1 (ISP gateway router)	GigabitEthernet0/1 (Internet)

```
  [CORE ROUTER]
           |
           ▼ (Out Interface: Virtual Serial0/0/0)
           |
   === Long Private WAN Wire ===
           |
           ▼ (Next Hop IP: 192.168.10.2)
      [MUMBAI EDGE ROUTER]
           |
           ▼ (LAN Interface)
=== Mumbai Branch Network (Destination: 10.1.0.0) ===
      [Local Mumbai Restaurant]
```